Nosh CV Intern Task — Report

Repo: github.com/maulik0jain/NoshInternTask · last updated 30 Oct 2026

Part	Status
Task 1 — ground-truth masks with a pretrained model	done
Dev / eval split (4:1, leakage-aware)	done, frozen
Task 2 — classical image-processing segmentation + experiment log	next
Task 3 — evaluation & analysis	after Task 2
Task 4 — video tracking (Drive link goes here)	after Task 3

0. Dataset — quick facts

- 110 images, all 640×640 (Roboflow resized with "stretch")
- Roboflow **folder** export → class labels only. **No boxes, no masks.**
- README says no augmentation → checked: 110 unique source ids, 0 copies
- 1–3 eggs per image, almost all white eggs
- Hard backgrounds: white bag, white egg trays, water in a steel pot, leaves, ash, hands, glossy lids

1. Task 1 — ground truth with pretrained models

Model choice

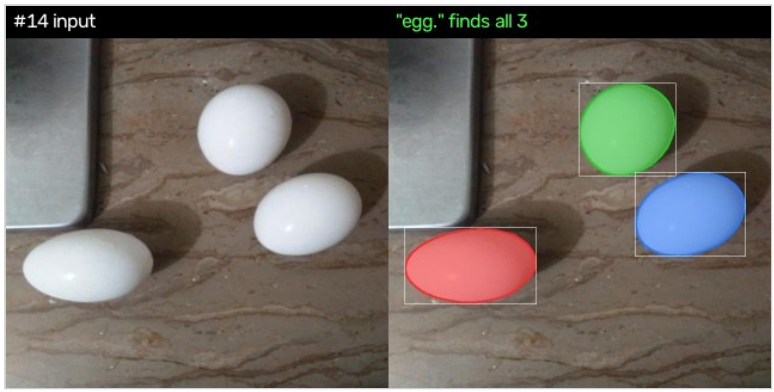
- SAM 2.1 needs a prompt (point / box) — it doesn't find eggs on its own
- Dataset has no boxes → **Grounding DINO** (open-vocabulary detector) turns the text `"egg."` into boxes
- Each box → **SAM 2.1 Hiera Base+** → mask
- Why not the lightest model: GT is made once, offline. Every GT mistake later counts as a mistake of my classical method → spend compute here
- Why not SAM "segment everything": it also segments hands, floor, shadows → messy filtering
- Also considered SAM 3 (text prompt built in) — heavier (~850M params) and gated weights; Grounding DINO + SAM 2.1 was enough

Pipeline

1. Grounding DINO base, prompt `"egg."`, box thr 0.30, text thr 0.25
2. Box cleanup: NMS (IoU 0.5), drop boxes >95% of the image, drop a "group" box that contains ≥2 other boxes
3. SAM 2.1, one mask per box (a box prompt isn't ambiguous → single-mask output)
4. Mask cleanup: clip to box (+6 px), fill holes (specular highlights), drop specks <15% of the biggest piece
5. Drop duplicate masks (>50% covered by a more confident egg)
6. Manual review overrides — 2 images, see below
7. Save binary mask, instance mask (ids left → right), overlay, `annotations.json`

Problems found while checking every overlay

Problem	Cause	Fix
Output files had wrong names	Filenames have extra dots (`x_jpg.rf.<hash>.jpg`), `with_suffix` cut off the hash	Build names from the stem
2 images missing one egg (#14, #35)	Prompt `"an egg."` gave those eggs 0.28–0.29 (threshold 0.30)	Prompt `"egg."`: real eggs 0.55–0.62, junk ≤ 0.10
Egg cut by a fridge-tray bar lost a piece (#24)	Cleanup kept only the largest piece	Keep all pieces ≥ 15% of the largest
Egg's reflection in a glossy lid detected as an egg (#61, #74)	Looks like a faint egg (det 0.36 / 0.47). Any rule for it would be fitted to 2 images	Manual override file `ground_truth_overrides.json`, with reasons
Not fixed: #17 — egg piece on the far side of a stem missing	SAM never segments that piece	Known GT error → will show up in Task 3



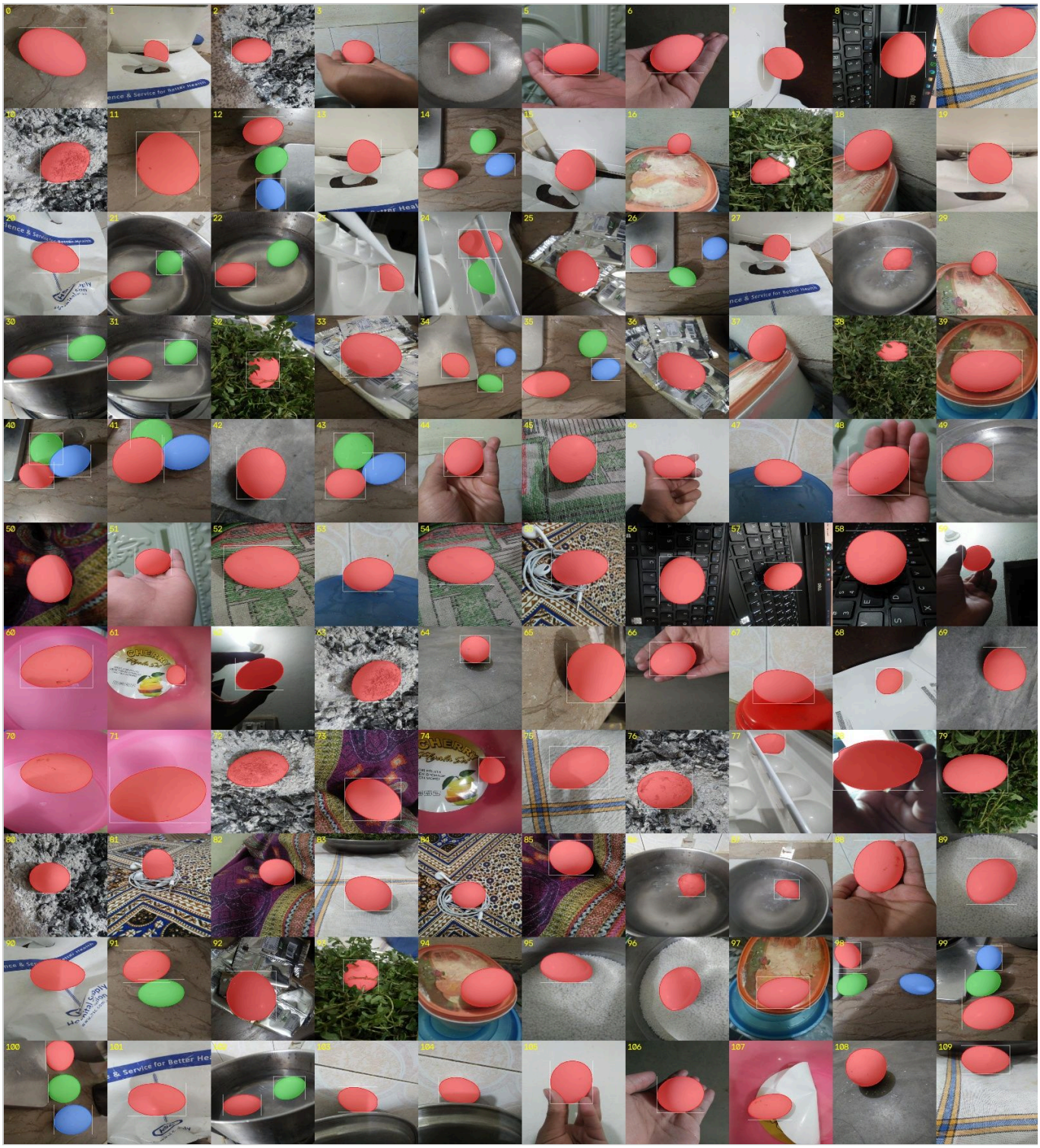
Prompt fix: all three eggs found with `"egg."`



Row 1: piece behind the bar kept. Row 2: reflection removed by override. Row 3: known GT limitation (not fixed).

Result

- **110 images, 139 eggs** — 92 images with 1 egg, 7 with 2, 11 with 3
- All 110 overlays checked by eye (full review sheet on the next page)
- GT = **visible** egg pixels. Leaves / fingers on top of an egg are not egg
- Known GT weak spots: eggs under leaves (#17), eggs partly hidden by tray bars



All 110 ground-truth overlays (colour = egg instance, white = detector box)

2. Dev / eval split (4:1)

Why a random split is wrong here

- Photos were shot in sessions: same pot, same keyboard, same bag, same hand — different angles
- Random split → same scene in dev and eval → tune thresholds on a scene, then "test" on the same scene → eval looks better than it is
- Same idea as splitting by subject, not by frame

Steps

1. **Augmented copies**: strip `.rf.<hash>` → source id. 110 ids, 0 duplicates
2. **Near-identical shots**: pHash (64-bit DCT), Hamming ≤ 10 → 1 pair, same group
3. **Same scene**: DINOv2-small embeddings (CLS + mean patch), average-linkage clustering, cosine distance 0.30 → 35 groups
   - 0.30 picked by looking at the groups. 0.40 merged marble floor with grey floor (19 images) and still split the pot
4. **Manual merges (8)**: same physical setup the embedding kept apart — pot, keyboard, fridge tray, white bag + holder, pink bowl, painted lid, grey floor, towel + rug → `scene_merges.json` → **19 groups**
5. **Split whole groups**. Rules: eval = 17–23% of images; multi-egg share within ±5 points of overall; then as many scenes as possible. Seeded search (seed 42)
   - Try 1: `StratifiedGroupKFold` → valid, but eval had only 4 scenes
   - Try 2: soft penalty → 7 scenes, but only 1 multi-egg image in eval (too easy) → made the multi-egg share a hard rule

Result

	dev	eval
Images	85	25
Eggs	109	30
Scenes (groups)	14	5
Images with 2+ eggs	15%	20%

- Eval scenes: steel pot (12, incl. 5 multi-egg), painted container (7), dark-room hand (3), blue bowl (2), red bowl (1)
- 85 : 25 = 77 / 23, not exactly 80 / 20 — whole groups don't divide evenly

Leakage check

For each eval image: cosine similarity (DINOv2) to its most similar dev image.

Split	Mean of max similarity	Eval images with a near-twin in dev (> 0.9)
Random split, same size	0.80	16%
Group split (used)	**0.59**	**0%**

- pHash near-duplicate pairs across dev / eval: 0

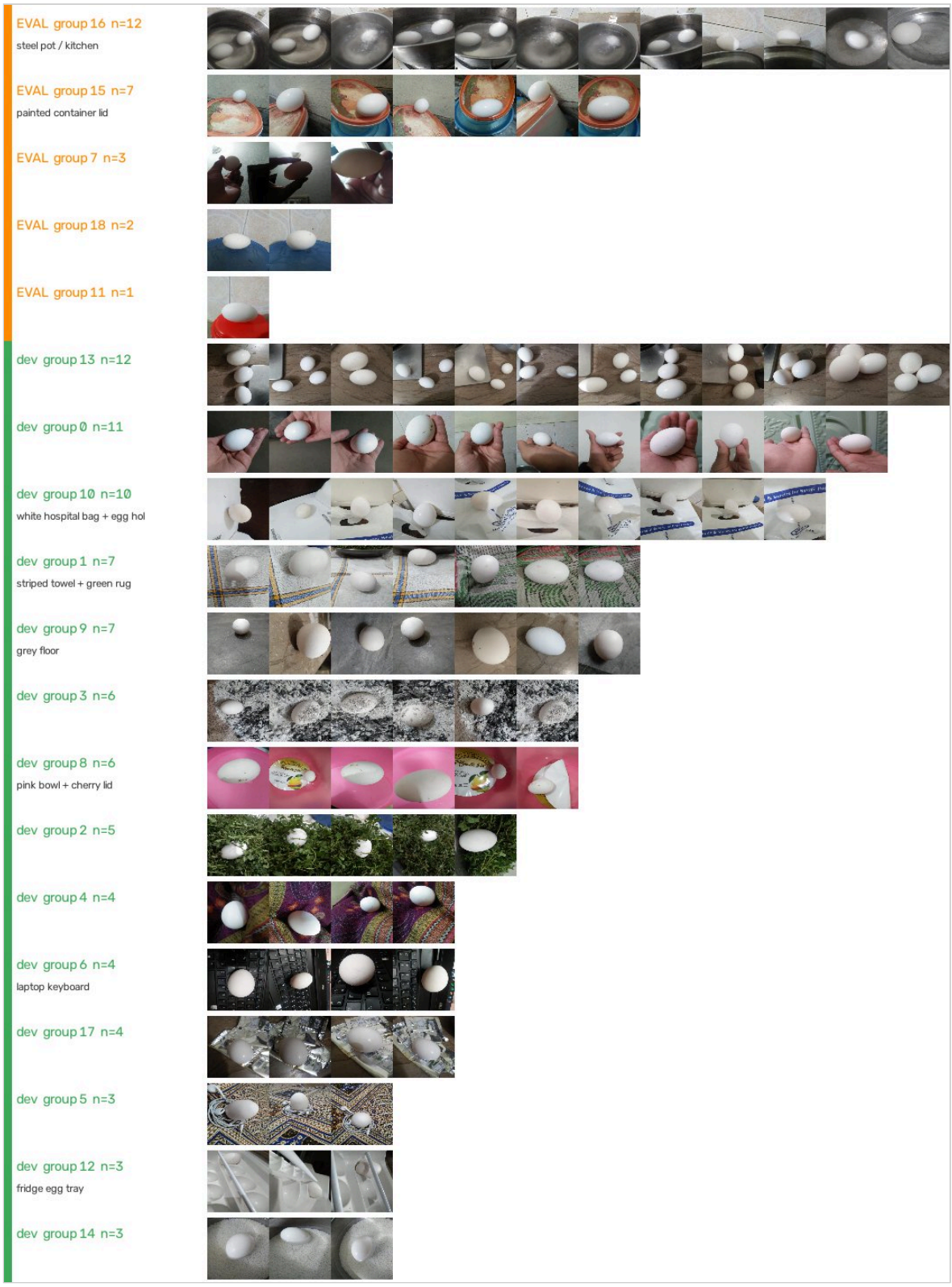
Rules from here on

- Split frozen in `data/splits/split.csv` (committed)

- All Task 2 tuning and experiments on **dev only**
- Eval touched **once**, at the end, for the Task 3 numbers

Limitations

- Scene grouping is partly manual (my judgement) — every merge is listed in a file
- Some scene types are only in dev (leaves, ash, bright-room hands, marble-floor multi-egg) → eval numbers say less about those



All 19 groups. Orange = eval, green = dev.

3. How to run

```
pip install -r requirements.txt
python scripts/download_assets.py          # SAM 2.1 checkpoint (+ dataset if missing)
python scripts/task1_sam2_groundtruth.py   # -> ground_truth/
python scripts/make_split.py               # -> data/splits/split.csv
```

- Tested on Windows 11, Python 3.13, PyTorch 2.6 + CUDA 12.4, RTX 4070 Laptop (8 GB)

4. AI tools

- Claude Code (AI coding assistant) used for: writing the scripts (GT pipeline, split, figures) and drafting this report